

Quantum Computing-Question Bank

1. Write the fundamental postulates of quantum mechanics
2. Write any four differences between classical and quantum computing.
3. What is a Linear vector space? Explain its axioms.
4. Explain Dirac Notation and explain how to represents Ket vector and Bra Vector in matrix form.
5. Explain Conjugate of a matrix and Transpose of a matrix.
6. Explain Pauli matrices and the interaction of Pauli matrices on 0 and 1 state.
7. Explain the interaction of Identity operator on 0 and 1 states.
8. Define an observable in quantum mechanics and give an example.
9. Explain how operators are used to represent observables in quantum mechanics.
10. Define eigenvalue and eigenvector in the context of linear algebra.
11. Explain how eigenvalues and eigenvectors are related to observables in quantum mechanics.
12. Define a Hermitian operator and explain its significance in quantum mechanics.
13. Explain how to calculate the expectation value of an observable using matrix representation.
14. Define a unitary operator and explain its role in quantum gates.
15. Show that the product of two unitary operators is also unitary.
16. What are Dirac notations and why is it useful in quantum mechanics?
17. Explain inner product with the bra-ket notations for two quantum states.
18. Explain the normalization and orthogonality properties of quantum states.
19. Define adjoint of an operator.
20. How are quantum states represented as column vectors in matrix notation?
21. Introduce the Pauli matrices and provide their matrix representations.
22. Explain Qubit and its properties
23. What is a qubit? Write any four properties of qubits.
24. Explain Single qubit and Two qubit
25. Explain Quantum superposition, Normalization, and Orthogonality of quantum states
26. Explain unitary Matrix and Pauli matrix
27. Write a note on Bloch Sphere with diagram.
28. Show $|\psi\rangle = |0\rangle$ and $|\psi\rangle = |1\rangle$ on the Bloch sphere.
29. Compare classical and Quantum gates
30. What are quantum gates and how are they different from classical logic gates?
31. Explain Hadamard Gate
32. Give an example of a quantum gate and explain its action on a qubit.
33. Explain Phase Gate (or S Gate) and T Gate.
34. Explain the Pauli X, Y, and Z gates with truth table

35. Explain the operation of a CNOT gate ?
36. Show how the Pauli matrices are related to basic quantum gates like the Hadamard and NOT gates.
37. Represent the qubits in matrix form.
38. Explain how to represent qubits in Bloch sphere.
39. Represent qubits and superposition states in Bloch sphere.
40. Explain the properties of quantum computing.
41. Apply identity Gate on quantum states 0 and 1 to show that there is no change.
42. Explain the quantum NOT gate. How it operates on qubits?
43. Explain Pauli-X gate. How it operates on qubits? Write the truth table for it.
44. Explain Pauli-Y gate. How it operates on qubits? Write the truth table for it.
45. Explain Pauli-Z gate. How it operates on qubits? Write the truth table for it.
46. Use matrix form of H to demonstrate how Hadamard gate operate on qubits. Write the truth table for it.
47. Explain S gate. How it operates on qubits? Write the truth table for it.
48. Explain T gate. How it operates on qubits? Write the truth table for it.
49. what operation does S gate perform on $\alpha|0\rangle + \beta|1\rangle$?

Numericals:

1. Verify if $|w\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $|v\rangle = \begin{pmatrix} 1 & -1 \end{pmatrix}$ are orthogonal.
2. $|\psi\rangle = A(2|0\rangle + 3i|1\rangle)$ find $\langle\psi|\psi\rangle$.
3. Show that (1,-1), (1,2) and (2,1) are linearly dependent.
4. Check for orthogonality $|i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$, $|-i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$.
5. Find if the following matrix is unitary:

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

6. Show that matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$ is unitary

$$7. |\psi\rangle = \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} 2 \\ -i \\ 2-3i \end{pmatrix}$$

find the inner product $\langle\phi|\psi\rangle$

8. Convert Ket vector to Bra vector:

$$|\Psi\rangle = \begin{pmatrix} 2-i \\ 3+2i \end{pmatrix}$$

9. Find Adjoint of Operator A

$$A = \begin{pmatrix} 2 & -i & 4 \\ 1-2i & 5 & 2-5i \\ -5i & 2+i & 3+7i \end{pmatrix}$$

10. Show that $A = \begin{bmatrix} 1 & 4+3i \\ 4-3i & 5 \end{bmatrix}$ is Hermitian